

SECOND EDITION

BREATHING AND RELAXATION

The Lost Innate Science



THE RELAXATION RESPONSE · RUSSIAN SYSTEMA

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Second edition · Free sample

Breathing and Relaxation: The Lost Innate Science
Second edition, 2026

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Health disclaimer. This book is for educational purposes only. It is not medical advice and does not replace diagnosis or treatment by a qualified health professional. The practices described here are intended to complement, never to replace, medical care. Consult your doctor before starting breathing or relaxation practices if you are pregnant, or if you have a heart, lung, blood-pressure, neurological or psychiatric condition. Stop any exercise that causes dizziness, pain or distress.

About the examples. The people described in the illustrative examples are composites created for teaching purposes; they do not describe real patients.

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Introduction

In a world that moves at breakneck speed, stress has become an inescapable part of life. From the pressures of work and relationships to the constant barrage of information, our bodies and minds are often in a state of overdrive. Chronic stress takes a toll on our health and is associated with a wide range of conditions, from hypertension and irritable bowel syndrome (IBS) to autoimmune disorders and mental health challenges.

But what if there were a simple, scientifically studied way to counteract the effects of stress? This is where the work of Dr. Herbert Benson comes in. In the 1970s, Dr. Benson, a Harvard-trained cardiologist, described the *Relaxation Response*: a natural, innate mechanism that counteracts the stress response and promotes deep relaxation. His research showed that practices such as meditation, prayer and controlled breathing could elicit this response, with measurable effects on the body.

At the heart of the Relaxation Response is breathing. Breathing is unique in that it is both an automatic and a voluntary function. By consciously changing the way we breathe, we can influence our nervous system and shift from the fight-or-flight response to a state of calm and balance. This connection between breath and relaxation is central to many traditional practices, including yoga, tai chi and the Russian martial art of Systema.

Systema, which translates as "the system," is a holistic approach to movement, breathing and relaxation. Developed in Russia, it emphasizes natural, effortless breathing as the foundation of physical and mental resilience. Its techniques are designed to reduce tension, sharpen focus and preserve calm even in high-stress situations.

This book explores the science behind the Relaxation Response and the art of Systema breathing. It offers practical tools to strengthen self-control, improve well-being and support people living with chronic diseases. Whether you want to manage stress, support your health or simply live a more balanced life, the practices in this book can help you tap into the power of calm. The book ends with a 7-day starter plan so that you can begin practicing today.

PART 1

**The Science of the Relaxation
Response**

CHAPTER 1

The Stress Epidemic and Its Impact on Health

The Physiology of Stress

What exactly happens inside our bodies when we experience stress? To understand this, we must look at the physiology of stress, and in particular at the fight-or-flight response, a mechanism that evolved to help us survive in the face of danger.

The Fight-or-Flight Response: A Survival Mechanism

The fight-or-flight response is an automatic physiological reaction to perceived threats, whether physical, emotional or psychological. It is orchestrated by the sympathetic nervous system, a branch of the autonomic nervous system that prepares the body to confront danger or flee from it. When the brain perceives a threat, it sends a distress signal to the hypothalamus, which activates the sympathetic nervous system.

This activation triggers a cascade of hormonal changes, most notably the release of adrenaline (epinephrine) and cortisol by the adrenal glands. Adrenaline increases heart rate, raises blood pressure and boosts energy by stimulating the breakdown of glycogen into glucose. Cortisol, often called the "stress hormone," increases the availability of glucose and temporarily suppresses functions that are not immediately needed in a life-threatening situation, such as digestion and parts of the immune response.

The Body in Overdrive: Short-Term versus Chronic Stress

In the short term, the fight-or-flight response is highly adaptive. It sharpens our senses, increases our strength and speed and prepares us to deal with immediate threats. Problems arise when this response is activated too often or for too long. Chronic stress, the persistent activation of the fight-or-flight response, can have lasting effects on the body.

When the body is repeatedly flooded with stress hormones, health can suffer. Sustained stress is associated with higher blood pressure, which strains the cardiovascular system. It is also linked to digestive complaints such as irritable bowel syndrome, and the long-term dysregulation of the immune system can make the body more susceptible to infections and may aggravate autoimmune disorders, in which the immune system attacks the body's own tissues.

The Role of the Amygdala and the Prefrontal Cortex

The fight-or-flight response is not only a hormonal reaction; it also involves key brain structures. The amygdala, an almond-shaped cluster of nuclei deep within the brain, plays a crucial role in processing emotions, particularly fear. When the amygdala detects a threat, it signals the hypothalamus, which activates the sympathetic nervous system.

The prefrontal cortex, the part of the brain responsible for decision-making and rational thought, helps modulate this response. Ideally, it regulates the amygdala and allows us to assess whether a perceived threat is real. Under chronic stress, however, this regulation can weaken, leading to exaggerated stress reactions.

The Vicious Cycle of Chronic Stress

Chronic stress creates a cycle that can be difficult to break. Research suggests that prolonged stress can affect brain structure and function, including the hippocampus, a region involved in memory and emotion regulation. This can make it harder to regulate emotions, which in turn feeds the stress response.

Chronic stress can also encourage unhelpful behaviors such as overeating, substance use and social withdrawal. These behaviors create new stressors and keep the cycle going.

Breaking the Cycle

Understanding the physiology of stress is the first step toward breaking this cycle. In the next chapter, we will explore Dr. Herbert Benson's discovery of the Relaxation Response, the biological opposite of the stress response. By learning to activate it, we can counteract the harmful effects of chronic stress and support our overall well-being.

CHAPTER 2

Dr. Herbert Benson's Discovery of the Relaxation Response

In the late 1960s and early 1970s, Dr. Herbert Benson, a cardiologist and researcher at Harvard Medical School, studied the physiological effects of meditation. He identified a natural, innate response of the body that he called the *Relaxation Response*, the biological opposite of the fight-or-flight response (Benson, Beary & Carol, 1974). His work provided a scientific framework for understanding relaxation, and practical tools for harnessing its benefits.

The Biological Opposite of the Stress Response

The fight-or-flight response, driven by the sympathetic nervous system, prepares the body to face immediate threats by increasing heart rate, blood pressure and stress hormones. The Relaxation Response, by contrast, is associated with the parasympathetic nervous system, which promotes calm and restoration. When it is activated, heart rate and breathing slow down, blood pressure decreases, muscles release their tension and the body's oxygen consumption drops.

Dr. Benson showed that the Relaxation Response is not merely the absence of stress but an active physiological state that can be deliberately elicited. This demonstrated that people could *actively* counteract the effects of chronic stress by engaging in specific practices.

The Key Components

Dr. Benson described simple elements that help elicit the Relaxation Response. They can be applied through many practices, including meditation, prayer, yoga, or even repetitive activities such as jogging or knitting. In his later work he emphasized the first two as the essential steps:

1. **Repetition.** Focusing on a word, sound, phrase, prayer, movement or on the breath, in a repetitive way. Repetition quiets the mind and shifts attention away from stressful thoughts. In meditation, a mantra (a repeated word or phrase) is often used to anchor the mind.
2. **A passive attitude.** Letting go of the need to control the process or to achieve a particular result. When distracting thoughts arise, you simply notice them, without judgment, and gently return to the repetition.
3. **Focus.** Directing attention to a single point: the breath, an image or the sensations of the body. Concentrating on one point breaks the chain of stress-inducing thoughts.
4. **A quiet environment.** Not always necessary, but minimizing distractions makes it easier to practice, especially at the beginning.

What the Research Shows

Dr. Benson's work was followed by decades of studies on relaxation and breathing practices. The main effects reported include:

1. **Lower heart rate and blood pressure.** By engaging the parasympathetic nervous system, relaxation practices counteract the cardiovascular effects of stress. This makes them a useful complement in the management of hypertension, always alongside medical follow-up.
2. **Immune function.** Chronic stress can dysregulate the immune system. Studies suggest that regular relaxation practice may help, mainly by reducing the stress load; the effects vary from person to person and remain an active area of research.
3. **Emotional well-being.** Relaxation practices have been shown to reduce symptoms of anxiety and stress, and can complement the treatment of mental health conditions.

4. **Better sleep.** By reducing mental and physical arousal, they help many people fall asleep more easily.
5. **Pain management.** By lowering muscle tension and promoting calm, relaxation can reduce the perception of pain and help people cope with chronic pain.

Practical Applications

The Relaxation Response is now widely recognized as a tool for stress management. Related approaches are used in programs such as mindfulness-based stress reduction (MBSR) and cognitive-behavioral therapy (CBT), and in clinical settings to support people living with chronic illnesses. It is accessible to everyone, regardless of age, fitness level or background, and requires no special equipment.

Try it: the classic Relaxation Response (10 minutes)

1. Sit comfortably, close your eyes and relax your muscles, from your feet to your face.
2. Breathe through your nose. Each time you breathe out, silently repeat a word you like (for example "calm," "one," or a short prayer).
3. When thoughts come, notice them without judgment and return to your word.
4. Continue for 10 to 20 minutes, then sit quietly for a minute before getting up.

In the next chapter, we will explore the central role of breathing in activating the Relaxation Response.

END OF THE FREE SAMPLE

Continue Reading

You have just read the introduction and the first two chapters of *Breathing and Relaxation: The Lost Innate Science*.

The full book continues with:

- The role of breathing in the Relaxation Response, with four practical techniques
- The Russian Systema approach: its four pillars, its breathing techniques and relaxation practices
- Three classic Systema drills
- Emotional regulation, support for people living with chronic diseases, and daily routines
- High-stress environments and the future of mind-body medicine
- A 7-day starter plan, safety guidelines and scientific references

Get the full book

Available on Amazon Kindle and on the Breathe & Relax website.